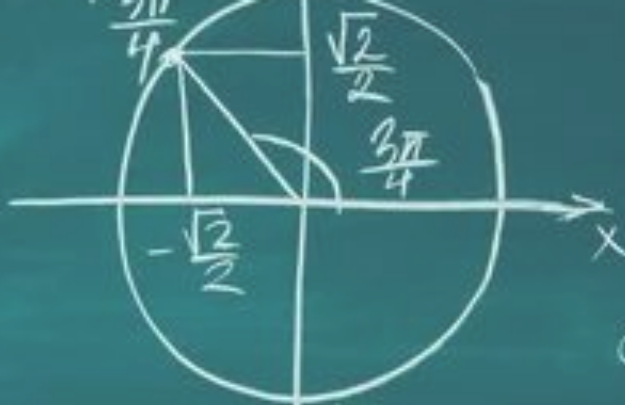
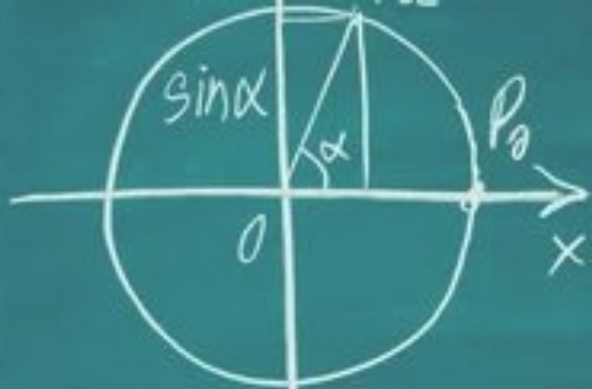




$$\operatorname{tg} \frac{3\pi}{4} = -1$$

$$\sin \frac{3\pi}{4} = \frac{\sqrt{2}}{2}, \quad \cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}, \quad \operatorname{tg} \frac{3\pi}{4} = -1$$

$$D(\sin) = D(\cos) = 0, \quad E(\sin) = E(\cos) = [1; 1]$$

First year Aerospace Engineering

Algebraic Solving exercises

Marcos Carrera Valle

Isolate x

- 1) $y = 0.25x^3 - 0.5$
- 2) $y = 3/2 * \sqrt{x + 4}$
- 3) $y = 5 / (2x - 1)$
- 4) $y = 100 * (0.8)^x$
- 5) $y = \ln(3x + 1/2)$
- 6) $y = 1/2 * m * x^2$
- 7) $y = 4 * e^{(0.1x)} - 2$
- 8) $y = 2 * \pi * \sqrt{x / 9.8}$
- 9) $y = (x^2 + 1.5)^{(1/3)}$
- 10) $y = 2 * \log_{10}(x / 5)$

Isolate x

11) $y = P / (1 + x)^n$

12) $y = 10 - 2 * \cos(3x)$

13) $y = 1 / (1 + e^{-x})$

14) $y = P * (1 + 0.04/12)^{(12x)}$

15) $y = \tan(\pi * x - 1)$

16) $y = 3 * 4^{(x/2)} + 1$

17) $y = \ln(\sqrt{x} / 3)$

18) $y = 0.5 * m * (x^2 - v^2)$

19) $y = (1 - x^{(1/4)})^2$

20) $y = m / \sqrt{1 - x^2/c^2}$

Find the roots of the following functions

21) $f(x) = x^2 - 7x + 12$

22) $f(x) = 2x^3 - 54$

23) $f(x) = e^{(2x)} - 3$

24) $f(x) = \ln(x + 2) - 1$

25) $f(x) = 2\sin(x) - 1$ (Solve for $0 \leq x < 2\pi$)

26) $f(x) = (3x + 1)/(x - 2) - 4$

27) $f(x) = \sqrt{2x + 3} - x$

28) $f(x) = 9^x - 27$

29) $f(x) = \cos(2x) + 1$ (Solve for $0 \leq x < 2\pi$)

30) $f(x) = x^4 - 81$

Find the roots of the following functions

31) $f(x) = x^2 - 6x + 8$

32) $f(x) = x^2 + 3x - 10$

33) $f(x) = 2x^2 - 5x - 3$

34) $f(x) = x^2 + 4x + 1$

35) $f(x) = x^2 - 12x + 26$

36) $f(x) = x^2 + 5x + 3$

37) $f(x) = 3x^2 - 2x - 4$

38) $f(x) = x^2 - 4x + 13$

39) $f(x) = x^3 - 2x^2 - 9x + 18$

40) $f(x) = x^4 - 10x^2 + 9$

Plot the following inequalities. Shade the region defined by the inequality

41) $y \leq x^2 - 4$

42) $y > 2\sin(x)$

43) $y \geq |x| - 3$

44) $x^2 + y^2 < 9$

45) $y < e^x - 2$

Solve the following system of linear equations

46)

$$2x + 3y = 12$$

$$x - y = 1$$

47)

$$x + y + z = 6$$

$$2x - y + z = 3$$

$$3x + 2y - z = 4$$

In a balanced ecosystem, the population of Rabbits (x) and Foxes (y) must satisfy two linear conditions to remain stable:

- 1) The total population of both species combined is 500.
- 2) To maintain the food chain balance, there must be 4 times as many Rabbits as there are Foxes.

48) Write the system of equations and solve for the number of Rabbits (x) and Foxes (y).

Consider a circuit with a voltage source and three nodes labeled A, B, and C.

Node A is connected to:

An 82V battery through a 4Ω resistor.

Ground (0V) through a 2Ω resistor.

Node B through a 4Ω resistor.

Node B is connected to:

Node A through a 4Ω resistor.

Ground (0V) through a 2Ω resistor.

Node C through a 4Ω resistor.

Node C is connected to:

Node B through a 4Ω resistor.

Ground (0V) through a 2Ω resistor.

49) Using Kirchhoff's Current Law (KCL) for each node, write down the system of 3 linear equations and solve for the voltages V_A , V_B , and V_C .

Aerodynamicists are analyzing the drag on a new experimental airfoil. The total drag is made of two parts: a shock-wave component that depends on the Mach index x (Shock wave drag = 2^x) and a constant parasitic drag (y).

Two wind tunnel tests gave the following results:

In the first test, 5 times the shock-wave drag plus 2 times the parasitic drag equaled a total force of 44 kN.

In the second test, 3 times the shock-wave factor minus 4 times the parasitic drag resulted in 0 kN

50) Write down the system of equations and solve for the Mach index and the parasitic drag y .

SOLUTIONS

Isolate x

- 1) $x = \text{cbrt}(4(y + 0.5))$
- 2) $x = (2y/3)^2 - 4$
- 3) $x = (5/y + 1) / 2$
- 4) $x = \log(y/100) / \log(0.8)$ (or $x = \log_{0.8}(y/100)$)
- 5) $x = (e^y - 1/2) / 3$
- 6) $x = \sqrt{2y / m}$
- 7) $x = 10 * \ln(y + 2) / 4$
- 8) $x = 9.8 * (y / (2\pi))^2$
- 9) $x = \sqrt{y^3 - 1.5}$
- 10) $x = 5 * 10^{(y/2)}$

Isolate x

11) $x = (P/y)^{(1/n)} - 1$

12) $x = (1/3) * \arccos((10 - y) / 2)$

13) $x = \ln(y / (1 - y))$

14) $x = \ln(y/P) / (12 * \ln(1 + 0.04/12))$

15) $x = (\arctan(y) + 1) / \pi$

16) $x = 2 * \log_4((y - 1) / 3)$

17) $x = 9 * e^{(2y)}$

18) $x = \sqrt{(2y/m) + v^2}$

19) $x = (1 - \sqrt{y})^4$

20) $x = c * \sqrt{1 - (m/y)^2}$

Find the roots of the following functions

21) $x = 3, x = 4$

22) $x = 3$

23) $x = 0.5 * \ln(3)$

24) $x = e - 2$

25) $x = \pi/6, x = 5\pi/6$

26) $x = 9$

27) $x = 3$ ($x = -1$ is an extraneous solution)

28) $x = 1.5$ (or $x = 3/2$)

29) $x = \pi/2, x = 3\pi/2$

30) $x = 3, x = -3, x = 3i, x = -3i.$

Find the roots of the following functions

31) $x = 2, x = 4$

32) $x = 2, x = -5$

33) $x = 3, x = -1/2$

34) $x = -2 \pm \sqrt{3}$

35) $x = 6 \pm \sqrt{10}$

36) $x = (-5 \pm \sqrt{13}) / 2$

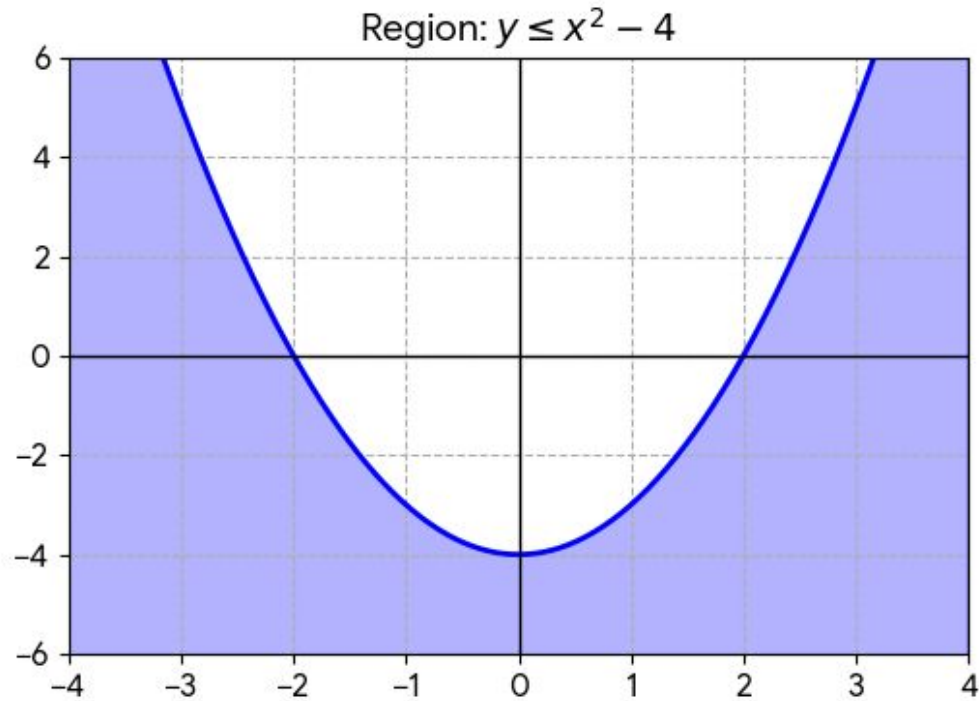
37) $x = (1 \pm \sqrt{13}) / 3$

38) $x = 2 \pm 3i$

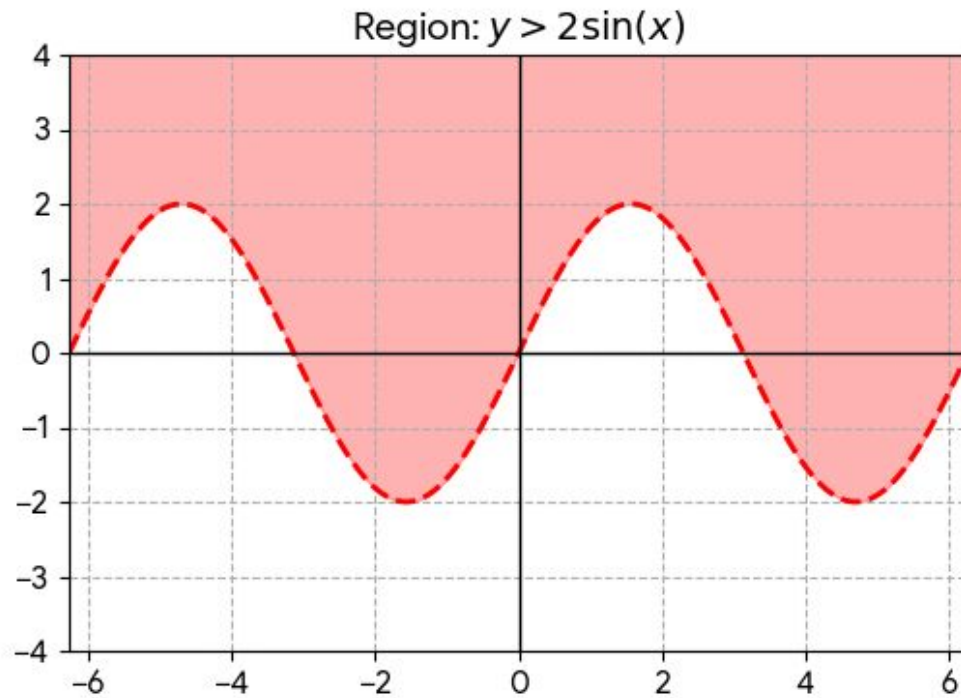
39) $x = 2, x = 3, x = -3$

40) $x = 1, x = -1, x = 3, x = -3$

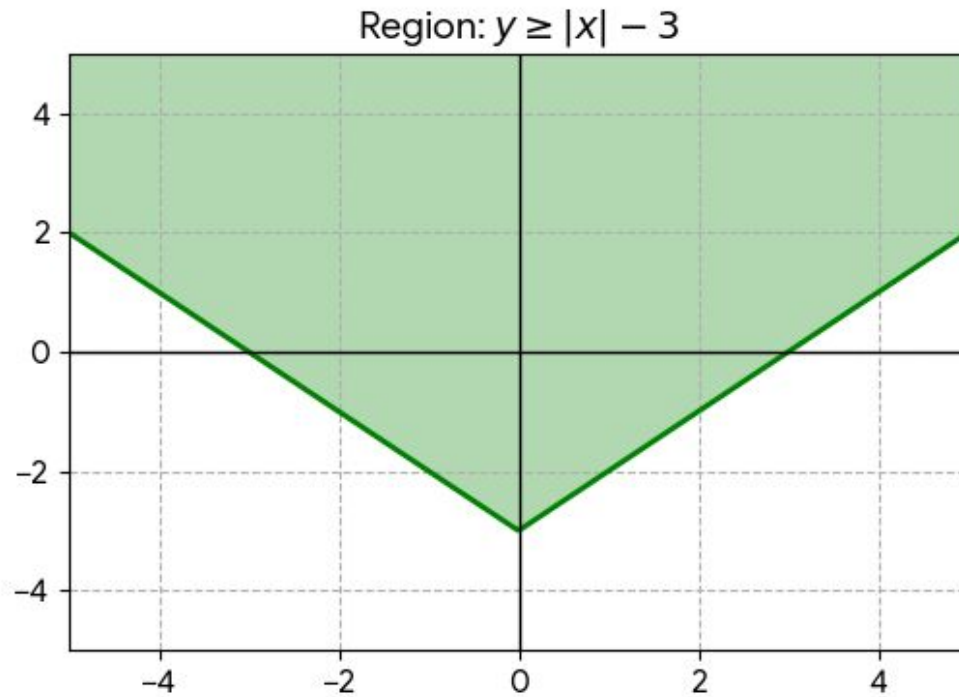
41)



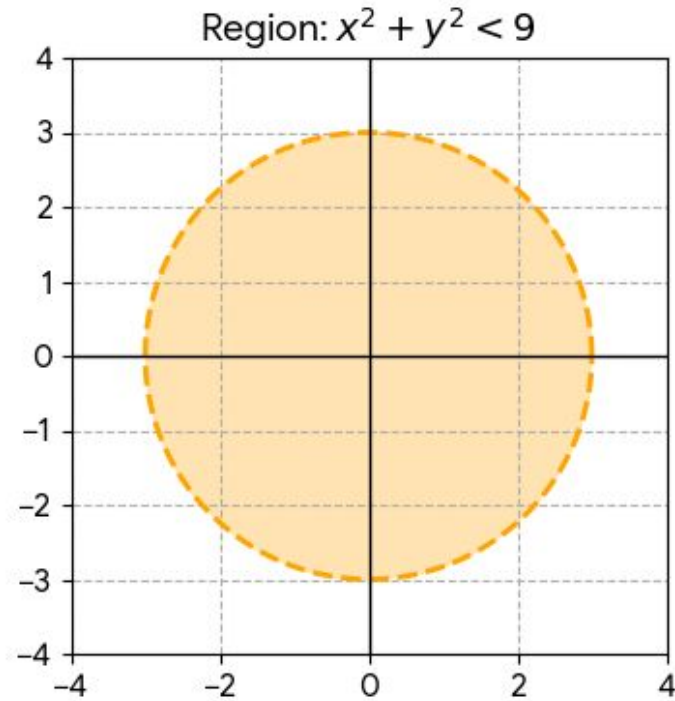
42)



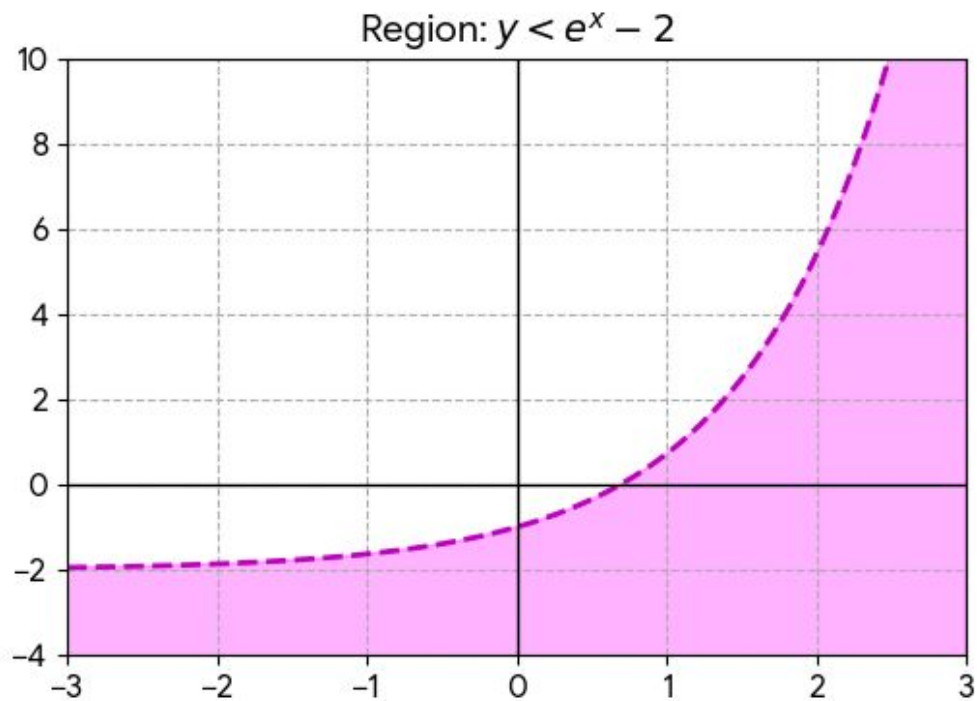
43)



44)



45)



Solve the following system of linear equations

46) $x = 3, y = 2$

47) $x = 1, y = 2, z = 3$

48) Write the system of equations and solve for the number of Rabbits (x) and Foxes (y).

$$x + y = 500$$

$$x = 4y \text{ (or } x - 4y = 0)$$

$$x = 400 \text{ (Rabbits)}$$

$$y = 100 \text{ (Foxes)}$$

49) Using Kirchhoff's Current Law (KCL) for each node, write down the system of 3 linear equations and solve for the voltages V_A , V_B , and V_C .

$$\text{Node A) } 4V_A - V_B = 82$$

$$\text{Node B) } -V_A + 4V_B - V_C = 0$$

$$\text{Node C) } -V_B + 3V_C = 0$$

$$V_A = 22V$$

$$V_B = 6V$$

$$V_C = 2V$$

50) Write down the system of equations and solve for the Mach index and the parasitic drag y .

$$5 \cdot 2^x + 2y = 44$$

$$3 \cdot 2^x - 4y = 0$$

$$x = 3$$

$$y = 6$$